CiteView: In-Situ Citation Sensemaking for Scholarly Reading

ANONYMOUS AUTHOR(S)

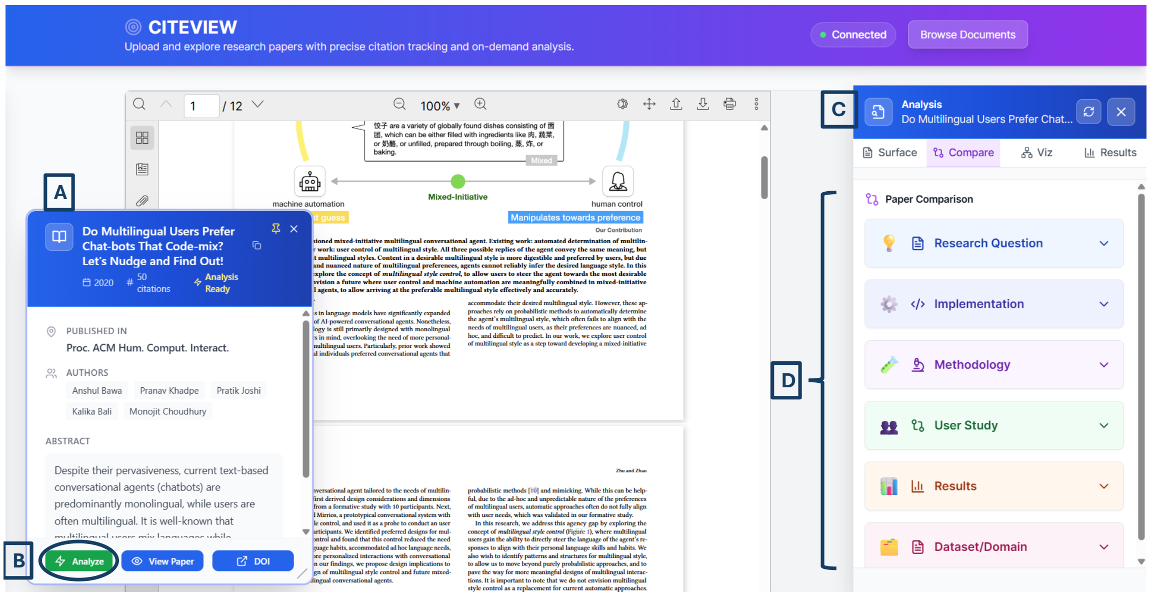

Fig. 1. CiteView interface overview. (A) Paper metadata card displays key bibliographic information. (B) The *Analyze* button initiates on-demand processing of the paper and its citations, triggering extraction of citation contexts, retrieval of cited papers, and generation of multi-modal analysis. (C) The Analysis sidebar opens when users click citations, presenting four complementary tabs: Surface, Compare, Viz, and Results. that respectively surface the most relevant context and citation reasons, contrast the citing and cited papers, visualize the cited work through figures and a semantic graph of its flow, and foreground its main empirical findings.(D) The Compare tab organizes side-by-side comparisons across six research dimensions, including Research Question, Implementation, Methodology, User Study, Results, and Dataset/Domain, each expandable to reveal detailed paired descriptions.

Academic papers rely extensively on citations, yet understanding the role and contributions of cited works often requires readers to interrupt their reading flow and consult multiple external sources. Moreover, this process imposes significant cognitive load. We present CiteView, an novel AI-assisted citation analysis tool that supports in-context exploration of cited papers directly within the PDF reading interface. When a user uploads a research paper, CiteView displays key metadata and enables citation analysis through an interactive Analyze function, which provides access to four complementary tabs. The Surface tab offers context-aware explanations of why a paper is cited at a specific location and how it is commonly used across the literature. The Compare tab enables side-by-side analysis across multiple research dimensions, such as methodology and user study design. The Viz tab generates semantic graphs that facilitate understanding the conceptual structure of the cited work and highlight key figures along with their relationships to the citing

Manuscript submitted to ACM

paper. The Results tab extracts and organizes cited paper findings—including quantitative metrics, user study outcomes, and empirical insights—to help readers assess their relevance to the citing paper. We evaluate CiteView through a mixed-methods laboratory study with twelve participants performing citation-focused reading tasks. We collect interaction logs, post-task questionnaires (SUS and NASA-TLX), and semi-structured interviews. Participants reported positive usability, low workload, and high efficiency. Qualitative feedback indicated reduced friction in accessing cited-paper contributions and a strong preference for insight-driven views that highlight results, enable direct comparisons, and support visual interpretation.

CCS Concepts: • **Human-centered computing → Human computer interaction (HCI)**; **Interactive systems and tools**; **User interface management systems**.

ACM Reference Format:

Anonymous Author(s). 2018. CiteView: In-Situ Citation Sensemaking for Scholarly Reading. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX).* ACM, New York, NY, USA, 21 pages. https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

Citations serve as the foundation of scholarly knowledge, linking new contributions to prior work and enabling readers to trace the evolution of ideas across research papers [10, 25]. Yet understanding how a citation supports an argument, whether through empirical evidence, methodological precedent, or contrasting findings, remains surprisingly difficult. Readers encountering a citation face a disruptive choice: accept the author's characterization of the cited work on faith, or break their reading flow to manually verify the claim by navigating to the source paper. This context-switching imposes significant cognitive load [29, 33] and leads readers to either skip verification, risking misinterpretation [2], or invest substantial time investigating references at the expense of deeper analysis [11]. In particular, with research papers citing many references (e.g., 20+), this friction makes thorough verification impractical.

Despite citations being fundamental to scholarly argumentation, existing reading tools offer limited support for accessing the specific context needed to interpret them [7, 16]. Existing tools address parts of this problem but not the core challenges, namely the fragmentation resulting from a mismatch between researchers' need to maintain reading continuity and the insufficient support for context-aware citation information revealing citation intent. Text-based assistants like SciSpace Copilot [26] clarify passages but do not reveal how cited papers support claims. Citation platforms such as VOSviewer [31] visualize co-citation and bibliographic coupling networks but cannot extract contextually relevant content, figures, results, or evidence from cited papers. Reference managers like Zotero, Mendeley, and EndNote[1] organize bibliographic metadata but offer limited support for understanding citation intent or retrieving visual evidence. These limitations are further exacerbated by the fact that researchers must navigate between disconnected systems to access information for a single citation. Therefore, it is of paramount importance to treat citations as integrated knowledge nodes that can be explored in-situ with multi-modal evidence (textual summaries, visual figures, empirical results, cross-paper comparisons), rather than simple references looked up elsewhere. An effective system should automatically surface what cited papers contribute, why they were cited, and what specific evidence supports claims, all without disrupting reading flow.

To that end, we developed CiteView (Figure 1), a system that integrates multi-modal citation analysis, textual summaries, visual evidence extraction, quantitative findings, and cross-paper comparison, directly within the article (PDF) reading interface. CiteView addresses three core challenges in citation verification: (1) eliminating context switching by surfacing multi-modal evidence (e.g., text and figures) from cited papers directly within the reading

[1] Zotero: https://www.zotero.org/; Mendeley: https://www.mendeley.com/; EndNote: https://endnote.com/

interface. (2) revealing citation intent by structured comparison between a cited work and the source paper across six key research dimensions (e.g., methodology)

(3) enabling on-demand analysis that processes user-uploaded PDFs directly through automatic extraction and retrieval, without requiring all papers to exist in pre-indexed databases.

When users upload a research paper, CiteView displays metadata (Figure 1A) and enables citation analysis via the Analyze button (Figure 1B). Clicking a citation opens an Analysis sidebar (Figure 1C) with four tabs: *Surface* provides context-aware explanations of why the paper was cited in this specific location and how it is typically used across the literature; *Compare* (Figure 1D) presents side-by-side analysis across six research dimensions including methodology, user study design, and implementation details; *Viz* generates semantic graphs showing the conceptual structure of the cited work and displays its key figures with explanatory captions; and *Results* extracts and organizes the cited paper's quantitative metrics and qualitative findings for rapid verification. This integrated interface enables exploration without breaking reading flow.

CiteView is evaluated through a mixed-methods study with 12 participants performing citation-focused reading tasks. The results show high *effectiveness*, with near-perfect task completion, low *cognitive load*, strong *usability*, and high ratings for *efficiency* and *ease of information discovery*, indicating that CiteView effectively reduces key challenges in citation exploration. Qualitative feedback showed that CiteView significantly improved the citation exploration experience, making citation checking a seamless part of reading and helping users quickly understand why a paper was cited and what evidence supports its claims.

This paper makes three primary contributions:

- We design CiteView, a system to integrate multi-modal, contribution-focused citation analysis directly within a unified PDF reading interface. Unlike prior work that treats citations as simple references or provides generic summaries, CiteView reveals what specific contributions cited papers make and why they were cited through multilevel analysis scheme.
- We introduce a four-level analysis framework, Surface, Compare, Viz, and Results, that uncovers the relationship between cited papers and their citing contexts, enabling a structured and multi-dimensional understanding of citations.
- We conduct comprehensive experiments to evaluate the system. Results demonstrate that the proposed system supports near-perfect task completion with low cognitive load, and participants report that CiteView makes citation checking a seamless part of the reading process.

In the rest of the paper, we review related work on interactive reading tools and citation analysis systems, present CiteView's design and implementation, report findings from our user study, and discuss design implications for citation-aware reading interfaces.

2 Related Works

Our work builds upon two intersecting areas: interactive tools for scholarly reading and citation analysis, and AI-assisted document understanding. Despite significant progress in each domain, there remains a lack of integrated support for contextual citation analysis during active reading.

2.1 Interactive Reading Tools and Citation Network Systems

Prior research on scholarly reading has consistently shown that academic reading is not a purely linear activity; instead, it involves iterative navigation, annotation, and cross-referencing as readers interpret and organize ideas[14]. Early systems like XLibris and LiquidText [27, 30] supported active reading through flexible navigation and spatial reorganization. Building on this, recent tools augment PDFs to reduce reading friction: ScholarPhi [9] surfaces just-in-time definitions, Paper Plain [3] adds plain-language summaries for non-experts, and Scim and PaperMage [7, 17] highlight key sentences and reflow layouts. While effective for local comprehension, these tools treat PDFs as isolated artifacts rather than entry points into broader evidence networks.

Research on sensemaking emphasizes that comprehension requires synthesizing relationships between a paper's content and external scholarly knowledge[19, 21]. Some recent systems integrate citation context into reading: PaperQuest [22] suggests relevant papers during literature review, CiteSee [5] augments inline citations with personalized relevance indicators, and CiteRead [23] annotates documents with commentary from subsequent works. Related efforts explore the discovery of the literature through author networks: ComLittee [12] enables scholars to curate committees of key authors to guide the recommendations of the articles, while Synergi [13] explores mixed-initiative workflows to synthesize research threads in multiple articles, helping scholars build hierarchical structures of related work. However, these systems primarily focus on incoming citations, author-based discovery, or text-based synthesis rather than analyzing outgoing references, extracting visual evidence, or enabling in-situ verification of specific cited claims within the active reading flow.

To navigate literature growth, numerous systems visualize citation networks and bibliometric patterns. Traditional tools like CiteSpace and VOSviewer [6, 31, 32] provide bibliometric overviews for identifying clusters and trends. Interactive platforms like Connected Papers, ResearchRabbit, Inciteful, and Litmaps[2] explore citation neighborhoods through co-citation relationships. While effective for discovery and landscape mapping, they operate exclusively on metadata and cannot surface specific content or figures that make citations relevant. These systems operate at the macro-level of citation networks rather than the micro-level of individual claims, showing connections between papers without explaining how specific cited content supports particular arguments. Most depend on pre-indexed databases [1] and cannot analyze user-provided PDFs on demand. Collectively, these tools leave readers without integrated support for verifying individual citations through direct access to the specific findings, figures, and contextual evidence that justify cited claims within the active reading flow.

2.2 AI-Assisted Document Understanding

Large language models have enabled AI-powered reading assistants that help researchers understand academic papers. Systems like SciSpace [26], Elicit [8], and Consensus [20] use generative AI to summarize content and answer questions. Smart citation platforms like scite.ai [24] classify citation statements as supporting, contrasting, or neutral. However, these tools typically treat documents as unstructured text and overlook visual elements like figures and tables, resulting in outputs that lack specificity for understanding how cited works support particular claims [15].

The Semantic Reader project integrates AI-driven context into reading interfaces by analyzing document structure [17]. Features like inline Citation Cards provide key details at the point of citation through in-situ annotations [5, 23]. Despite this progress, these systems rely on indexed corpora, cannot process arbitrary user-uploaded PDFs, and offer

[2]https://www.connectedpapers.com/, https://www.researchrabbit.ai/, https://inciteful.xyz/, https://www.litmaps.com/

limited support for extracting figure-level evidence, primarily facilitating textual understanding without integrated multimodal support for citation intent [17, 24].

Inspired by CitePeek's approach to surfacing contextual paragraphs through similarity scoring [4], CiteView extends in-document citation analysis by integrating comparative, visual, and multi-dimensional analytical capabilities. This reflects the recognition that citation comprehension requires understanding not only what a cited work contributes, but also how it relates to the current paper, what visual evidence it provides, and how its findings connect across analytical dimensions.

In summary, prior work has advanced important components of the scholarly workflow: active reading tools support annotation, citation network systems enable discovery, and AI-driven services provide summarization and semantic insight [6, 18, 28, 31, 32]. However, these capabilities remain fragmented. Reading tools rarely integrate external sources, visualization platforms operate away from the moment of reading, and AI assistants function outside the natural flow of PDF-based engagement. Even citation-aware reading systems do not deliver concrete evidence from cited papers directly into the reading view.

This fragmentation motivates the need for an integrated, in-document approach. CiteView addresses this gap by treating the PDF as an entry point to a connected scholarly knowledge graph rather than a static endpoint. Unlike tools that rely on context switching or pre-indexed databases [6, 17, 24], CiteView operates on-demand on arbitrary PDFs to surface citation intent, comparative relationships, and visual evidence from referenced figures or tables directly alongside the cited text, enabling efficient verification and deeper cross-document sensemaking without disrupting the reader's flow.

3 System Overview

CiteView (Figure 1) transforms static PDFs into citation-centric exploration environments. Clicking a citation opens a metadata card (Figure 1A) with three buttons: *Analyze*, *View Paper*, and *DOI*. Clicking *Analyze* (Figure 1B) opens an Analysis sidebar (Figure 1C) with four views, Surface, Compare, Viz, and Results, providing contextual summaries, cross-paper comparisons, visualizations, and key findings directly within the reading interface.

3.1 Surface Tab

The Surface tab is the default view in the Analysis sidebar, providing a fast, context-aware explanation of what the cited paper contributes in that specific place in the source document. It is designed not to summarize the entire article, but to answer a narrower question: "Why is this paper being cited here, and what does it add?" To achieve this, it combines the local citation context in the source PDF with selectively extracted content from the cited paper and transforms it into three to four concise, evidence-grounded bullet explanation points that can be read at a glance. (Figure 2)

To generate these context-aware explanations, CiteView employs a multi-stage pipeline. When a user clicks an inline citation, the system captures the complete sentence containing the citation (typically 15-40 words) and treats it as a query/context encoding the topic and rhetorical role/purpose of the citation (e.g., introducing a method, supporting a result). The cited paper is processed using GROBID[3] for metadata extraction and PyMuPDF for full-text extraction. Text is segmented into sentences, which are then ranked by relevance using a multi-component algorithm combining semantic similarity, keyword overlap, and section importance. This approach captures both lexical and semantic relationships while prioritizing content from methodologically important sections (e.g., Methods, Results). The top-ranked sentences

[3]GROBID: https://github.com/kermitt2/grobid

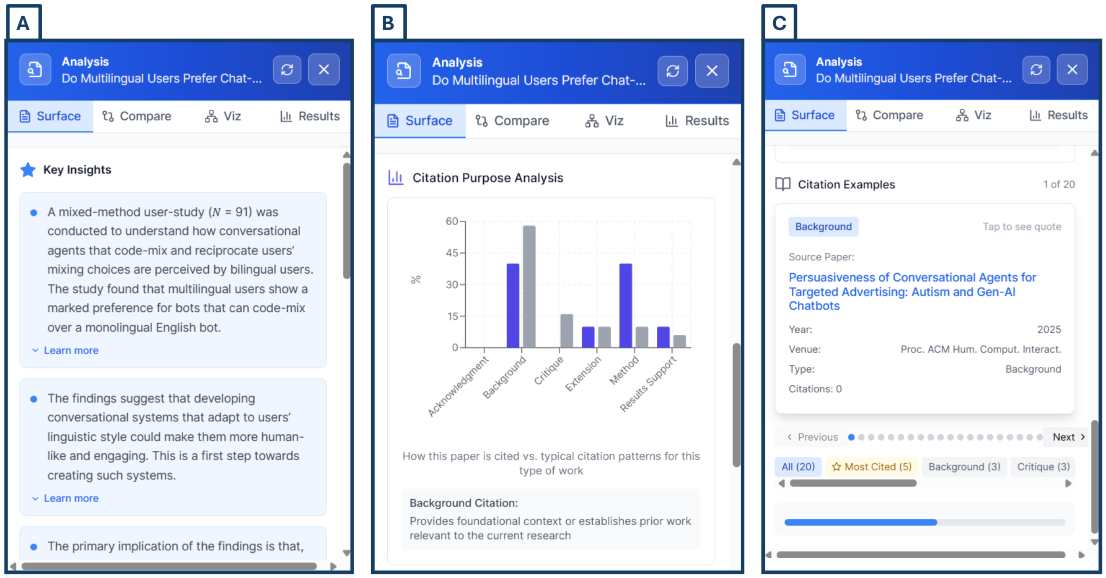

Fig. 2. Surface tab views in CiteView. (A) shows context-aware bullet points that explain the cited paper's role in the current citation, (B) visualizes citation-purpose distributions, and (C) lists example citing papers that use the same work, each card revealing the exact quotation when clicked.

(typically 5-8) are selected for analysis. The citation context and top-ranked sentences (300-600 words) are passed to OpenAI's GPT-4[4]. Structured prompts instruct the model to generate exactly 4 bullet points that explain the cited work's specific role and tie it to the local context, emphasizing concrete methods or results rather than generic summaries.

In the interface, the Surface tab presents these bullets alongside the original citation context, allowing readers to see both the author's sentence and the system's interpretation in one place (Figure 2.A). Each bullet is initially displayed in a compact form, with controls that allow users to expand it for full text when more detail is needed, supporting both quick scanning and in-depth inspection.

Moreover, the Surface view displays citation-purpose distributions showing how the cited work is typically used in the literature (Figure 2.B). The blue bar shows the current citation's purpose, classified by analyzing the citation context using GPT-4 to identify its rhetorical role (Background, Method, Results Support, Extension, Critique, Acknowledgement). The gray bars show typical citation patterns, derived by retrieving citation contexts from Semantic Scholar[5] and classifying them using keyword-based pattern matching across six categories (Background, Method, Results Support, Extension, Critique, Acknowledgement). These distributions reveal patterns in how the cited work is used across the literature. For example, the gray bars might show a paper is cited primarily for its methodology (60%) with few critique citations (5%), while the blue bar shows whether the current paper follows this typical pattern or uses it differently. The view also presents example citations from other papers retrieved from Semantic Scholar, each annotated with citation-purpose labels and basic metadata (title, venue, year). Selecting a citation card reveals the exact quoted passage, enabling readers

[4]OpenAI GPT-4: https://platform.openai.com/docs/models/gpt-4
[5]Semantic Scholar: https://www.semanticscholar.org/

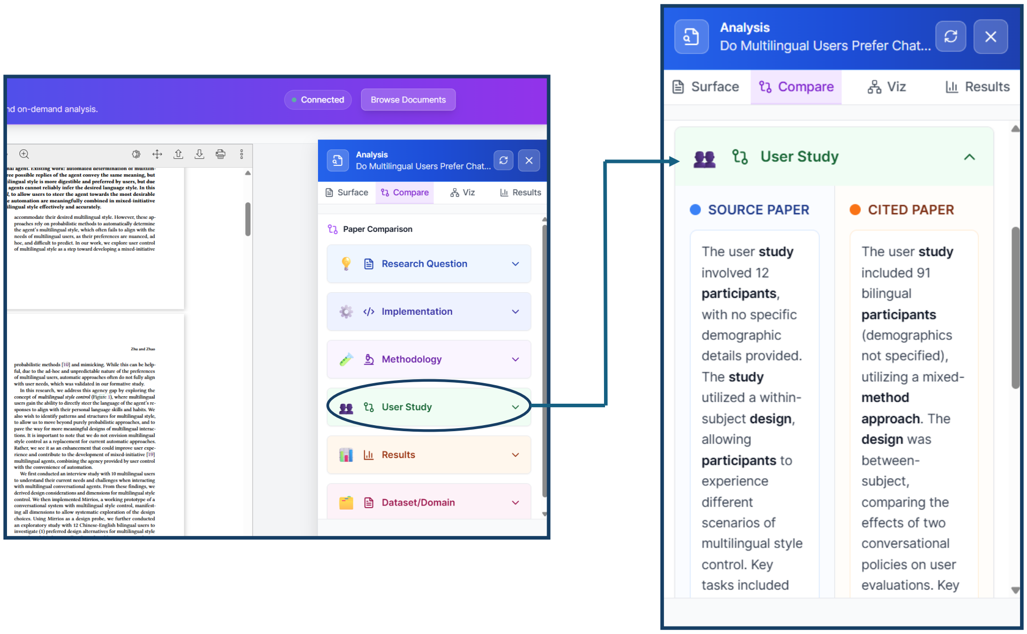

Fig. 3. Compare tab in CiteView. The left panel shows the main PDF reader with the Compare tab selected, and the zoomed view highlights a user-study card comparing the source paper (left column) and the cited paper (right column) along participants, design, and other study details.

to quickly understand how the cited paper is used in context, whether as background, methodology, or comparative results (Figure 2.C).

3.2 Compare Tab

The Compare tab enables readers to understand how a cited work relates to the source paper across six research dimensions: research question, methodology, implementation, user study, results, and dataset/domain (Figure 3).

These six dimensions address the core questions readers typically ask when evaluating cited work: what problem does it address (research question), how does it approach the problem (methodology), what system was built (implementation), how was it evaluated (user study), what was found (results), and where does it apply (dataset/domain). These categories emerged from iterative design reflecting how researchers structure empirical work in HCI, spanning the complete arc from problem framing through solution development to validation and contextualization. The dimensions balance comprehensive coverage with cognitive manageability. We limited the structure to six rather than more granular categories based on pilot testing with two graduate students, which indicated that more than six dimensions fragmented comparisons into overly specific units while requiring excessive scrolling.

When content is missing for a dimension (e.g., a theoretical paper lacks implementation, or a survey paper has no original user study), the system explicitly indicates absence rather than generating speculative content, displaying messages such as "Implementation not available, theoretical paper" or "User study not conducted, evaluation based on benchmarks." When dimensions are fundamentally incomparable (e.g., comparing a survey paper's literature review methodology with an experimental paper's controlled study), each paper's approach is described independently to avoid misleading parallels.

When a reader opens the Compare tab, the system employs a two-agent workflow, a multi-step AI pipeline where specialized components handle extraction and analysis, built with the CrewAI framework[6]. An *Ingestion Agent* first extracts structured content from both papers by identifying relevant sections (e.g., "Methods," "Results," "Implementation") through header detection and keyword matching, then capturing key information such as technical terms, system components, evaluation metrics, and experimental designs. This agent utilizes GPT-4 with dimension-specific prompts that specify the information to extract for each comparison dimension. An *Analyst Agent* then receives this structured content and generates six paired descriptions, one summarizing the source paper's approach, one for the cited paper, written in plain language with concrete details rather than generic statements.

Comparisons are displayed as collapsible side-by-side cards (source left, cited right), allowing readers to expand specific dimensions for detailed examination or scan across all six at once. This design transforms the manual process of switching between PDFs into a structured comparative overview that supports higher-level reasoning about the literature, such as identifying complementary methods, spotting opportunities for extension, or detecting when citations are methodologically misaligned with the work that cites them (Figure 3).

3.3 Viz Tab

The Visualization tab addresses a fundamental challenge: readers often need to quickly verify specific claims or understand a paper's core contribution without reading the entire document. Yet determining what problem a paper addresses, what approach it proposes, and what results it achieves requires navigating information scattered across multiple sections. The tab provides two complementary visual representations. The Semantic Graph sub-tab displays key concepts as an interactive node-link diagram, where nodes represent research elements (problems, approaches, findings). This enables readers to quickly grasp the paper's conceptual structure and contribution flow. For example, understanding how a specific problem motivated a particular approach that produced certain findings. The Figure sub-tab extracts and displays key figures from the cited paper with their captions, providing direct access to visual evidence without opening the cited PDF. Together, these views move beyond linear text summaries by exposing the conceptual structure of a paper and its visual evidence, enabling readers to reason about cited work more efficiently within the reading context.

3.3.1 Semantic Graph. The Semantic Graph view generates a compact semantic knowledge graph capturing the cited paper's argumentative structure (Figure 4). CiteView extracts and segments the paper's text, then processes each section using GPT-4 with section-specific prompts to extract concise statements as typed nodes. We selected eight node types, Problem, Approach, System, Finding, Dataset, Insight, Background, and Critique, to cover the essential elements of research contributions while maintaining visual clarity. These categories emerged from an analysis of how authors structure arguments in empirical papers, including identifying problems, proposing approaches, implementing systems, reporting findings, describing datasets, offering insights, providing background, and critiquing limitations. Nodes are connected by typed edges (e.g., "motivates," "implements," "produces") inferred through two complementary methods: (1) rule-based heuristics that apply predefined relationship patterns (e.g., Problem nodes connect to Approach nodes with "motivates" edges, Approach nodes connect to System nodes with "implements" edges, System nodes connect to Finding nodes with "produces" edges), and (2) semantic similarity scoring between node text pairs nodes with cosine similarity

[6]CrewAI: https://www.crewai.com/

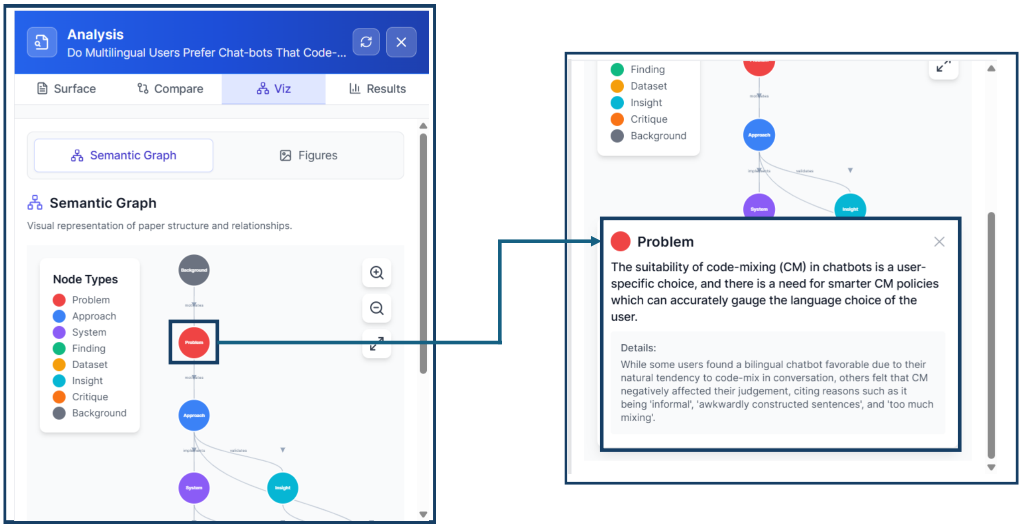

Fig. 4. Semantic Graph view showing the conceptual structure of a cited paper. Clicking on a node (e.g., Problem or Approach) opens a detailed explanation, supporting overview-to-detail exploration without leaving the reading context.

above a 0.6 threshold receive "relates to" edges. This hybrid approach balances structural precision (rule-based) with flexibility to capture unexpected relationships (similarity-based). When readers click a node, CiteView displays a focused textual explanation with supporting details, allowing for an overview-to-detail exploration without requiring linear reading.

3.3.2 Figures. The Figure view surfaces three curated figures from the cited paper: a teaser figure providing a high-level overview, the most important figure reflecting the paper's primary contribution, and the most related figure relevant to the current citation context (Figure 5). This three-figure selection emerged from balancing competing needs: readers require both broad understanding (teaser, important) and citation-specific evidence (relevant), but showing all figures (often 6-10 per paper) would overwhelm the interface and require excessive scrolling. Pilot testing with two graduate students confirmed that three figures provided adequate coverage, participants could locate relevant visual evidence without being overwhelmed, while showing only one figure frequently missed key contributions. CiteView extracts figures using PyMuPDF [7] to detect embedded image objects and identifies captions by searching for figure references (e.g., "Figure 1") within proximity thresholds of image boundaries. Figures are ranked by: (1) position in document, and (2) semantic similarity between figure caption and citation context (computed using sentence embeddings). The top-ranked figure from each category (position-based, context-relevance) is selected to provide diverse coverage.

Each figure is accompanied by a three-part explanation generated by GPT-4. We designed this structure to address the challenge that figures in academic papers are often difficult to interpret without reading surrounding context. Readers may see a system diagram or results graph but not understand its significance or how to read it. The three-part format

[7] PyMuPDF: https://pymupdf.readthedocs.io/

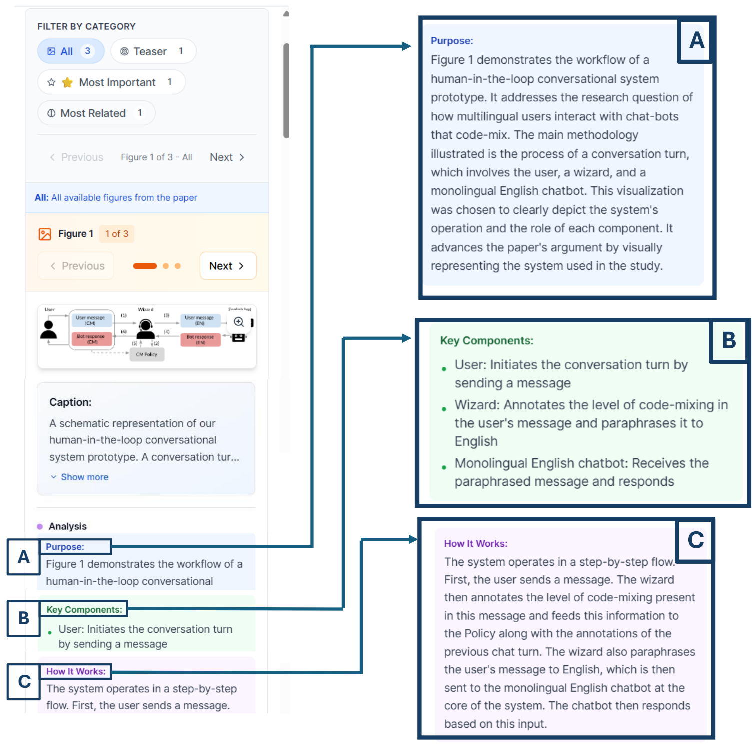

Fig. 5. Figure view in CiteView showing an extracted figure from a cited paper with structured explanations. (A) describes the figure's purpose and research question, (B) lists key components and their functions, and (C) provides a step-by-step explanation of how the depicted system works. Navigation controls allow browsing multiple figures (1 of 3 shown).

(1) Purpose describes the figure's role and research question (Figure 5A), (2) Key Components lists main visual elements (Figure 5B), and (3) How It Works explains how elements interact (Figure 5C), follows cognitive principles of figure comprehension (readers need purpose, then components, then mechanism). The prompt instructs GPT-4 to analyze the figure caption and surrounding text, generating these sections with specific constraints: Purpose (2-3 sentences), Key Components (3-5 bulleted items), and How It Works (sequential explanation). This enables figure interpretation without opening the cited PDF.

3.4 Result Tab

The Results tab addresses a fundamental challenge: readers need to quickly grasp cited paper findings, key metrics, user study outcomes, or empirical insights, to assess its relevance to their work. Manually extracting results requires navigating dense sections that contain multiple tables, figures, and scattered findings throughout the text. The Results tab automates this by extracting and structuring findings through three complementary views (Figure 6). These three views cover the essential types of information readers need from results sections: what was measured (quantitative), what was observed (qualitative), and why it matters to the current citation (relation). The Quantitative view extracts numerical outcomes as structured units containing measured metrics (e.g., "accuracy"), reported value (e.g., "85%"), comparison contexts (e.g., "vs. baseline: 78%"), and statistical significance (e.g., "$p < 0.05$"), enabling readers to quickly

Fig. 6. Results tab showing quantitative, qualitative, and relation views that summarize cited paper findings and explain their relevance to the current paper, supporting efficient interpretation without leaving the reading context. (A) presents quantitative outcomes as structured units with metrics, values, and statistical significance, (B) surfaces qualitative themes and insights with supporting evidence.

understand performance achievements. The Qualitative view identifies non-numerical findings such as user behaviors, interview themes, or design insights, presenting each as a concise claim paired with supporting evidence from the paper. This addresses the challenge that qualitative findings are scattered throughout results and discussion sections without standardized formats. The Relation view explains how cited results connect to the source paper through four types: validation (cited results support source claims), comparison (cited results provide a performance baseline), extension (source builds upon cited results), and critique (source identifies limitations in cited results). Content is extracted using three specialized CrewAI agents, each equipped with GPT-4 and domain-specific prompts: the Quantitative Analysis Agent for numerical findings, the Qualitative Analysis Agent for themes and insights, and the Relation Analysis Agent for cross-paper connections. Together, these three views provide comprehensive coverage of empirical findings, enabling readers to rapidly assess the cited paper's contributions and relevance without manually searching through dense results sections.

4 System Implementation

CiteView is implemented as a full-stack web system that integrates structured document processing, interactive PDF rendering, external scholarly metadata services, and LLM-based analysis. The frontend is built with React and TypeScript; the backend uses Python with Flask. The system combines rule-based extraction methods with LLM-based synthesis to balance reliability and expressiveness.

CiteView processes PDFs through a multi-stage pipeline. We render documents using PDF.js for interactive display, then extract citations using a combination of GROBID (for bibliographic structure) and PyMuPDF (for coordinate detection). This dual approach is necessary because GROBID provides rich metadata but no positional information, while PyMuPDF provides locations but limited semantics. Metadata enrichment is performed via the Semantic Scholar API, with SQLite caching to reduce redundant queries. When full-text access is required, the system queries open-access endpoints in sequence: Semantic Scholar, Unpaywall[8], arXiv[9], and PubMed Central[10], validating and caching retrieved PDFs. If no accessible version is found, CiteView degrades gracefully to metadata-only analysis.

CiteView employs retrieval-augmented generation (RAG): relevant content is extracted from cited papers, then provided to GPT-4 for grounded analysis. Text is segmented into sentences, which are then ranked using a multi-component scoring algorithm: (1) **Semantic similarity** computes cosine similarity between the citation context and each sentence using sentence-transformers embeddings (all-MiniLM-L6-v2), capturing conceptual alignment beyond exact word matches. (2) **Keyword overlap** measures shared technical terms using TF-IDF weighting to preserve domain-specific vocabulary. (3) **Section importance** assigns higher scores to sentences from the Introduction, Methods, and Results sections. (4) **Position weighting** favors sentences appearing early in their sections. These scores are combined, and the top 8 sentences above a relevance threshold are selected and passed to GPT-4 with structured prompts. For Surface analysis, the system passes the citation context and top-ranked sentences to GPT-4 with prompts instructing the model to generate exactly 4 bullet points (15-25 words each) that: (1) explain the cited work's specific role in the argument, (2) tie that role explicitly to the local citation context, and (3) emphasize concrete methods or results rather than generic summaries. The prompts explicitly prohibit including author names, citation numbers, or meta-commentary about the analysis process. Citation purpose classification analyzes the citation context using GPT-4 to categorize citations into six types: Background, Method, Results Support, Extension, Critique, and Acknowledgement. The system passes the complete sentence containing the citation to GPT-4 with a structured prompt requesting classification. For citation usage distribution across the literature, CiteView retrieves citation contexts from Semantic Scholar and applies keyword-based pattern matching to classify how other papers cite the same work. For the Compare and Results tabs, prompts enforce structured JSON output. To minimize hallucination, outputs are constrained through evidence-grounding (operating only on retrieved sentences), structured formats, external validation (Semantic Scholar citation labels), explicit refusal instructions, and rule-based fallbacks. Compare and Results tabs employ a multi-agent system built with CrewAI. The Compare tab uses two agents: an Ingestion Agent that extracts structured content from both papers by identifying relevant sections (Methods, Results, Implementation), and an Analyst Agent that generates six paired descriptions across research dimensions. The Results tab uses three specialized agents for quantitative, qualitative, and relational analysis. All agents use GPT-4 with dimension-specific prompts and output structured JSON validated against predefined schemas.

Figures are extracted using PyMuPDF and ranked by position, and semantic similarity to citation context. Each figure includes a GPT-4-generated explanation with three parts: Purpose, Key Components, and How It Works. Semantic graphs extract typed statements (Problem, Approach, System, Finding, Dataset, Insight, Background, Critique) using GPT-4, limited to 5-10 nodes. Relationships are inferred through rule-based heuristics and semantic similarity, then rendered using D3.js force-directed layout.

[8]Unpaywall: https://unpaywall.org/
[9]arXiv: https://arxiv.org/
[10]PubMed Central: https://www.ncbi.nlm.nih.gov/pmc/

5 User Study

To evaluate how CiteView supports citation sensemaking during academic reading, we conducted a user study examining its impact on reading flow, citation comprehension, and users' ability to interpret cited work without leaving the source paper. The study protocol was approved by Ontario Tech University's Research Ethics Board through the Integrated Research and Innovation System (IRIS).

5.1 Participants

We recruited 12 participants (4 women, 8 men) aged 23–26 (M=24.5), including 10 master's students and 2 senior undergraduates. Ten were from computer science, and two were from physics and management. All noted that they routinely read academic papers as part of their research or coursework and rely on citations to interpret related work. Participants were recruited through email invitations sent to university mailing lists. As an incentive, three participants were randomly selected to receive a $20 gift card for their time.

5.2 Study Design and Procedure

After reading the consent form and receiving an overview of the study, we introduced CiteView in detail, describing the system's purpose and the function of each tab (Surface, Compare, Results, and Visualization). Participants were then given time to explore the interface and interaction flow using a sample research paper. Once they were familiar and confident with the system, we briefed them on the tasks they would complete.

We selected three recent HCI research papers that cited multiple related works, each containing 30–50 citations with diverse citation purposes (background, methodology, results support). To test CiteView's reliability across different paper types, we employed a between-subjects design to avoid learning effects and fatigue from analyzing multiple papers: participants were randomly divided into three groups of four, with each group working with a different source paper throughout all three task blocks. This design ensured that our findings reflected CiteView's performance across diverse papers rather than being specific to a single document, demonstrating the system's generalizability across different HCI research contexts. For each paper, we pre-selected specific citations that were well-supported by CiteView's analysis capabilities, meaning the cited papers were openly accessible, contained clear empirical results, included figures, and could be meaningfully compared to the source paper across multiple dimensions. This selection ensured that all tabs would contain substantive information for participants to explore.

Each participant completed three task blocks designed to gradually build familiarity with CiteView.

In Block 1, participants were assigned a source paper and instructed to click on a specific citation (e.g., "[15]") within that paper. They were then explicitly told to use only the Surface and Compare tabs to answer four multiple-choice questions, two questions per tab. Questions for the Surface tab focused on citation context and purpose (e.g., "why is this paper cited in this context?" or "How is this work typically cited in the literature?"). Questions for the Compare tab focused on cross-paper relationships (e.g., "How does the cited paper's user study design differ from the source paper's approach?" or "How does the cited paper's implementation design differ from the source paper's approach?"). This block served two purposes: (1) it introduced participants to CiteView's ability to surface citation-specific context (Surface) and comparative analysis (Compare), and (2) it established that different tabs serve different analytical purposes. By restricting tab usage, we ensured participants engaged deeply with these two views before moving to visualization-oriented tabs.

Block 2 used the same citation from Block 1 but required participants to switch to the Results and Visualization tabs to answer four new multiple-choice questions, two per tab. Questions for the Results tab focused on empirical findings (e.g., "What were the key quantitative results reported in the cited paper?" or "What qualitative themes emerged from the user study?"). Questions for the Visualization tab focused on visual evidence and paper structure (e.g., "What does the main figure in the cited paper illustrate?" or "According to the semantic graph, what problem does the cited paper address?"). By reusing the same citation across blocks, participants could compare how Surface/Compare views complemented Results/Visualization views for understanding a single cited work. This design reinforced that CiteView provides multi-faceted analysis rather than redundant summaries. After completing Block 2, participants had now used all four tabs in a guided manner.

Block 3 assessed whether participants could independently select appropriate tabs to answer citation questions. Participants were assigned a different citation from the same source paper and given four multiple-choice questions without any restrictions on tab usage. Questions were intentionally varied to require different types of information (e.g., one question might require Results data, another might require Compare analysis, and another might require Visualization evidence). Participants had to determine which tabs contained the necessary information and navigate accordingly. This block served as an ecological validity check: after guided exposure in Blocks 1-2, could participants use CiteView effectively in a more realistic scenario where they must decide how to explore citations on their own? We recorded which tabs participants used for each question through interaction logs, allowing us to observe naturalistic tab selection patterns.

All multiple-choice questions were generated manually through a two-step process. First, we used CiteView to analyze each pre-selected citation across all four tabs. Second, we formulated questions whose answers could be directly verified from the displayed content. For example, if the Compare tab showed "User Study: Source paper used 12 participants; Cited paper used 8 participants," we created the question "How many participants were in the cited paper's user study?". This ensured questions were answerable using CiteView without requiring external knowledge or subjective interpretation. Each question was piloted during our preliminary testing to confirm clarity and answer correctness.

After completing all three task blocks, participants filled out the NASA-TLX workload questionnaire and the System Usability Scale (SUS). They then participated in a 20-minute semi-structured interview where we asked them to reflect on their experience, describe which features they found most useful, identify any points of confusion, and compare CiteView to their typical citation-following workflows. The entire study session lasted approximately 60 minutes. The procedure was validated through pilot testing with two graduate students, resulting in minor clarifications to task instructions.

5.3 Measures and Data Collection

We collected task performance (percentage correct), system logs (time per tab), NASA-TLX workload ratings, SUS usability scores, and semi-structured interview data. Quantitative measures were analyzed using descriptive statistics, while interview transcripts were thematically coded to identify recurring patterns.

6 Results

We present results from our user study that reports data from interaction logs, standardized questionnaires (NASA-TLX and SUS), and post-task semi-structured interviews. Participants successfully completed citation-focused tasks using CiteView and generally reported positive experiences with the system. Usability scores were high and perceived

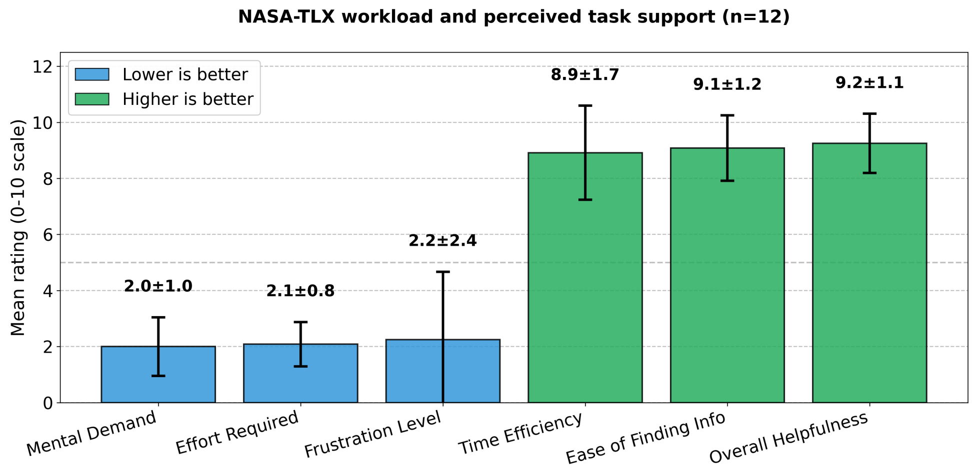

Fig. 7. NASA-TLX workload and perceived task support ratings (n=12, 0-10 scale). Error bars represent standard deviation. Lower scores indicate reduced cognitive and physical workload; higher scores indicate better efficiency and support. CiteView demonstrated low workload demands and high perceived usefulness across all dimensions.

workload was low, indicating that the system effectively supports citation-centric exploration. Qualitative feedback further highlighted the value of inline citation exploration while revealing concrete areas for refinement.

6.1 Task Performance

Participants completed the citation-focused questions with consistently high accuracy across all study sessions. Each participant answered 12 questions per session. Across all participants ($n = 12$), the mean score was 11.7 correct answers out of 12 (SD=0.62; median= 12; range: 10–12), with ten participants answering all questions correctly. Incorrect or missing responses were rare, typically limited to one or two questions per participant, and occurred on specific question types (e.g., questions asking participants to identify a paper's main figure). These errors were not associated with difficulties using the interface, and no recurring usability or interface-related problems were observed during task execution.

6.2 Usability and Workload

SUS results indicate clearly positive usability for CiteView (M=71.9, SD=13.2, n=12), exceeding the standard benchmark of 68 and nearing the commonly used "good usability" cutoff (73.5). Question-level responses reveal a highly polarized pattern consistent with well-designed systems: all positively worded items received strong ratings, while negatively worded items were rated low.

Participants reported high intention to use the system (Q1: M=4.25, SD=0.45), strong ease of use (Q3: M=4.33, SD=0.98), coherent integration of features (Q5: M=4.50, SD=0.52), easy learnability (Q7: M=4.17, SD=0.94), and confidence while interacting (Q9: M=4.33, SD=0.49). In contrast, negatively framed items were consistently rated low, with unnecessary complexity (Q2: M=2.08, SD=1.24), need for technical support (Q4: M=2.25, SD=1.42), inconsistency (Q6: M=1.67, SD=0.65), and difficulty of use (Q8: M=1.83, SD=0.94) all clustering near the desirable lower end of the scale. This

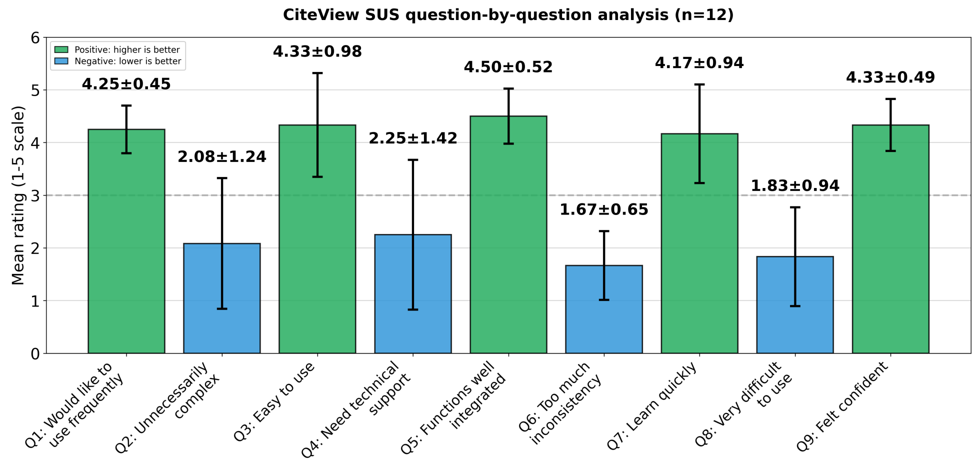

Fig. 8. Item-level SUS ratings for CiteView (n=12, 1-5 scale). Mean ratings with standard deviation error bars. Green bars represent positively worded items (odd-numbered: higher scores indicate better usability); blue bars represent negatively worded items (even-numbered: lower scores indicate better usability). The dashed line indicates the neutral midpoint (3.0). Results show strong agreement on positive items (M=4.25-4.50) and disagreement on negative items (M=1.67-2.25), indicating favorable usability perceptions.

divergence suggests that users experienced CiteView as straightforward, cohesive, and learnable, with only isolated friction points rather than systemic usability barriers (Figure 8).

NASA-TLX workload ratings confirm that participants experienced CiteView as low-burden and highly supportive during citation exploration. Mental demand (M=2.0, SD=1.0) and effort required (M=2.1, SD=0.8) were both at the low end of the 0–10 scale, indicating that interacting with the system did not feel cognitively demanding. Frustration levels were somewhat more variable (M=2.2, SD=2.4), suggesting that while most participants encountered little friction, a few experienced occasional confusion or slowdowns. In contrast, the positively oriented workload items were rated very highly: time efficiency (M=8.9, SD=1.7), ease of finding information (M=9.1, SD=1.2), and overall helpfulness (M=9.2, SD=1.1). This pattern, low demand and frustration paired with strong perceived support, indicates that CiteView enables fast, efficient navigation of citation information with minimal effort (Figure 7).

6.3 Qualitative Themes

We analyzed post-task semi-structured interviews using an inductive thematic analysis approach. We conducted open coding on the interview transcripts, followed by iterative axial coding to consolidate recurring patterns related to citation sensemaking, interface learnability, and perceived usefulness of CiteView's analytical views. Coding continued until no new conceptual categories emerged, indicating thematic saturation. Across participants, three central themes emerged: (1) eliminated context-switching and access barriers, (2) preference for comparison- and insight-oriented views, and (3) complementary features supporting diverse analytical needs.

Theme 1: Eliminated context-switching and access barriers. Participants consistently emphasized that CiteView substantially reduced the overhead traditionally associated with verifying citations. Rather than opening external links, locating PDFs, navigating paywalls, or scanning lengthy documents for relevant content, CiteView allowed them to access contribution-level information at the moment of need. P1 noted that the tool "made the information more readily available and easily accessible compared to looking for that individual PDF," describing the insights as available "instantaneously" and "in a short and concise way." P4 estimated that CiteView "reduced it by 10X," referring to the time required to check citations.

This reduction in friction was especially clear when participants compared CiteView with their typical workflow. P3 explained that without CiteView they would need to "go on Google, figure out if I had access to that paper," then authenticate through their institution and confirm that the retrieved paper matched the citation, steps eliminated entirely when "I just have to click the citation." P2 similarly stated that they could "read more papers within a short period of time," indicating that CiteView supported lightweight verification that would otherwise be too time-consuming. These comments reinforce that CiteView's in-situ, contribution-focused design directly supported reading continuity and eliminated context-switching costs.

Theme 2: Preference for comparison-, insight-, and visualization-driven views. Across participants, insight-oriented views, especially *Compare*, *Results*, and *Viz*, were consistently rated as the most valuable. Rather than wanting high-level summaries, participants preferred views that allowed them to understand how the cited paper differed from or related to the source paper, what results it produced, and what visual evidence supported these findings.

P1 identified the *Results* tab as their favorite because they could "instantly understand what the cited paper did." P2 highlighted the *Compare* tab, noting that the side-by-side structure made it easier to understand methodological differences and similarities. P3 emphasized "the graph and statistics tab" because it provided key insights "really quickly" and foregrounded "the important graphs or figures."

Participants also appreciated the citation-purpose visualization, with P6 calling it "a very smart decision" because it reveals "what this cited paper was used for the most," helping situate each citation within the broader literature. Interaction logs supported these perceptions: participants frequently navigated toward insight- and relationship-oriented views (e.g., *Compare*, *Viz*, *Results*) when answering citation-related questions. Collectively, these findings indicate that participants valued analytical representations that surfaced relationships, evidence, and contribution-level differences rather than descriptive summaries.

Theme 3: Complementary features supporting diverse analytical needs. A notable finding was that multiple participants explicitly valued the complete set of analytical views, viewing each tab as serving distinct but complementary purposes. Five participants (P6-P10) stated that all current features are "crucial and relevant," with P6 emphasizing they would "add more tools rather than getting rid of any of it" because "all of those tools are integral parts of research." This perception of feature completeness was reinforced by participants' varied preferences for different tabs based on their analytical goals: some prioritized Results for understanding contributions (P1, P4), others emphasized Compare for cross-paper relationships (P2), and still others highlighted Viz for rapid visual insights (P3).

Participants' task performance supported this view of complementary functionality. Interaction logs showed that different questions naturally led participants to different tabs, with no single view dominating usage. When given unrestricted tab access in Block 3, participants strategically selected appropriate views based on question type, demonstrating that the multi-view structure supported flexible, task-driven exploration rather than creating unnecessary redundancy. P2 explicitly requested additional analytical dimensions: "You can add more features, like a dataset, and then comparison of the accuracy in a table or something," indicating that participants saw value in comprehensive analytical coverage rather than interface minimalism. While feature completeness was valued, several participants suggested presentation refinements, for example, enhancing navigation (P6 requested keyword search functionality). These suggestions focused on interface clarity rather than removing analytical capabilities.

Overall, results demonstrate that CiteView effectively supports citation exploration through eliminating context-switching, analytical depth across multiple views, and complementary features addressing diverse sensemaking needs. Quantitative metrics (97.9% task completion, SUS 71.9, low cognitive load) and qualitative feedback converge on a system that successfully integrates multi-modal citation analysis into the reading workflow while maintaining usability. Refinement opportunities center on presentation clarity and progressive disclosure rather than fundamental design changes.

7 Discussion

Our results indicate that integrating citation analysis directly into the reading workflow enables fast and effective citation sensemaking without imposing substantial cognitive burden. Participants achieved near-perfect task performance while reporting low mental demand and effort, alongside high ratings for time efficiency, ease of finding information, and overall helpfulness. These outcomes suggest that surfacing detailed information in-situ can meaningfully reduce the overhead of citation verification while preserving reading flow, particularly when supported by comparison- and visualization-oriented views.

Across both quantitative measures and qualitative feedback, participants consistently preferred insight-oriented representations over purely descriptive summaries. Views that foregrounded results, cross-paper relationships, and visual evidence (i.e., Results, Compare, and Visualization) were perceived as most effective for understanding why a paper was cited and what it contributed. Participants' preferences reveal a clear design principle: citation tools should prioritize contribution-level analysis and multi-modal evidence over general summaries, with descriptive context serving as a lightweight entry point to deeper analytical exploration.

Results demonstrated that CiteView's analytical depth was successfully integrated into a usable interface, with positive usability ratings (SUS M=71.9) and remarkably low cognitive load (Mental Demand M=2.0, Effort M=2.1). Participant suggestions for refinement focused on presentation rather than functionality: visual legends and enhanced navigation aids. These refinements would further reduce the initial learning curve while preserving the comprehensive analytical coverage that participants explicitly valued, as evidenced by five participants stating all features were "crucial and relevant."

Based on these findings, we derive three design implications for citation exploration tools:

(1) Prioritize Insight-Driven Views While Retaining Descriptive Context. Participants strongly preferred views that supported interpretation and comparison over high-level descriptive summaries. Tabs emphasizing empirical results, methodological differences, and visual evidence enabled participants to quickly understand what a cited paper contributed and how it related to the source paper. As P1 noted, *"the results would be the most favourite because I can instantly understand what the cited paper did."* This preference reveals that readers engage with citations through specific analytical goals rather than seeking comprehensive summaries. Citation tools should therefore foreground contribution-focused analysis (results, comparisons, visual evidence) while providing descriptive context as a secondary orientation layer. Future interfaces might organize views by analytical purpose (e.g., "Verify Claims," "Compare Methods," "Find Evidence") rather than content type, better aligning with readers' goal-driven exploration patterns.

(2) Balance Feature Richness Through Progressive Disclosure. Participants appreciated CiteView's comprehensive analytical capabilities, including multi-dimensional comparisons, curated figures, and structured results. However, several noted that the breadth of features required initial familiarization, with P5 describing the design as *"amazingly designed"* while noting it was possible to *"get lost with so many features."* Successful elements such as collapsible comparison dimensions demonstrate the value of progressive disclosure, allowing readers to scan high-level insights before selectively expanding details. However, the four-tab structure required upfront understanding of each tab's role, contributing to early learning curves. Future designs should adopt hierarchical information architectures that surface core insights first, with clear, visually guided pathways to deeper analysis, preserving analytical power while reducing initial cognitive load.

(3) Support Question-Driven Navigation in Addition to Structured Exploration. While the tab-based structure effectively guided exploratory browsing, participants expressed a desire for more direct, question-driven access mechanisms. As P6 suggested, *"If I could just search within CiteView, it would tell me a paragraph... If there is any figure about it, it gives you an option to go to figures."* This feedback reflects the dual nature of citation use: readers alternate between open-ended exploration (e.g., gaining an overview of a cited work) and targeted fact-finding (e.g., checking a specific metric or method). Citation tools should therefore support both modes by combining structured overviews with in-tool search and contextual navigation, enabling readers to efficiently locate relevant evidence based on their immediate information needs.

Together, these implications suggest that effective citation exploration tools should foreground key insights, manage complexity through progressive disclosure, and flexibly support both exploratory and question-driven access patterns. Grounded in both performance outcomes and user feedback, these principles highlight how citation interfaces can transform citation checking from a disruptive detour into an integrated, low-friction component of scholarly reading.

8 Limitations and Future Work

Our study has several practical limitations. First, CiteView depends on access to the full text of cited papers; however, not all references are openly available due to paywalls, missing PDFs, or incomplete metadata. As a result, some citations cannot be expanded into summaries, comparisons, or figure retrieval, which limits coverage and can introduce inconsistency across papers. Second, document quality and formatting remain a constraint: older or non-standard PDF layouts (e.g., scanned documents, uncommon templates, or poorly structured text layers) may reduce extraction reliability and may not be fully supported by the current pipeline, affecting citation detection, section parsing, and figure linking.

These limitations also point to clearer, limitation-driven opportunities for future work. A first direction is improving robustness to missing or restricted-access references by integrating fallback strategies, such as metadata-only summaries, citation-network context, or partial extraction from abstracts, so users still receive meaningful insight even when full PDFs are unavailable. A second direction is strengthening support for irregular or low-quality documents by expanding the parsing pipeline (e.g., OCR, layout-aware models, and adaptive figure detection) to ensure more consistent extraction across diverse formats. Finally, given participant feedback about cognitive load and interface fragmentation, future versions should focus on consolidating views, unifying navigation, and adopting progressive-disclosure patterns that surface advanced analyses only when needed.

9 Conclusion

We introduced CiteView, a reading interface designed to bring evidence-level context from cited papers directly into scholarly workflows. By integrating summaries, relational views, and visual explanations into the PDF itself, CiteView aims to make citation exploration more immediate, more coherent, and less interruptive. Our study indicates that providing in-situ access to methodological details, findings, and visual evidence can help readers maintain focus and answer citation-driven questions with greater fluidity. Beyond the specific features we tested, the results point toward a broader direction: citation tools should act as pathways for understanding by revealing why a citation matters in the moment it becomes relevant.

Looking ahead, this work highlights the opportunity to reimagine citation interaction more broadly. Embedding richer, structured representations within reading environments may support new forms of comparative analysis, rapid sensemaking, and cross-paper reasoning. We see CiteView as a step toward more integrated, insight-oriented scholarly tools, and believe this direction can help transform citation follow-up from a disruptive task into a seamless part of academic reading and discovery.

References

[1] Waleed Ammar et al. 2018. Construction of the Literature Graph in Semantic Scholar. In *Proceedings of NAACL-HLT*. 84–91. doi:10.18653/v1/N18-3011

[2] Marc H. Anderson and Russell K. Lemken. 2023. Citation Context Analysis as a Method for Conducting Rigorous and Impactful Literature Reviews. *Organizational Research Methods* 26, 1 (January 2023), 45–73. doi:10.1177/1094428120969905

[3] Tal August, Lucy Lu Wang, Jonathan Bragg, Marti A. Hearst, Andrew Head, and Kyle Lo. 2023. Paper Plain: Making Medical Research Papers Approachable to Healthcare Consumers with Natural Language Processing. *ACM Trans. Comput.-Hum. Interact.* 30, 5, Article 74 (Sept. 2023), 38 pages. doi:10.1145/3589955

[4] Yasaman A. Basti, Heidar Davoudi, and Ali Neshati. 2025. CitePeek: Contextual Citation Exploration Within Research Papers. In *Proceedings of the 7th ACM Conference on Conversational User Interfaces (CUI '25)*. Association for Computing Machinery, New York, NY, USA, Article 20, 7 pages. doi:10.1145/3719160.3737639

[5] Joseph Chang, Amy Zhang, Jonathan Bragg, Andrew Head, Kyle Lo, Doug Downey, and Daniel Weld. 2023. CiteSee: Augmenting Citations in Scientific Papers with Persistent and Personalized Historical Context. 1–15. doi:10.1145/3544548.3580847

[6] Chaomei Chen. 2006. CiteSpace II: Detecting and Visualizing Emerging Trends and Transient Patterns in Scientific Literature. *Journal of the American Society for Information Science and Technology* 57, 3 (2006), 359–377. doi:10.1002/asi.20317

[7] Raymond Fok, Hita Kambhamettu, Luca Soldaini, Jonathan Bragg, Kyle Lo, Marti Hearst, Andrew Head, and Daniel S Weld. 2023. Scim: Intelligent Skimming Support for Scientific Papers. In *Proceedings of the 28th International Conference on Intelligent User Interfaces* (Sydney, NSW, Australia) *(IUI '23)*. Association for Computing Machinery, New York, NY, USA, 476–490. doi:10.1145/3581641.3584034

[8] Timothy Harrison. 2023. Elicit. *Journal of the Medical Library Association* 111, 2 (2023), 343–346. doi:10.5195/jmla.2023.1613

[9] Andrew Head, Kyle Lo, Dongyeop Kang, Raymond Fok, Sam Skjonsberg, Daniel S. Weld, and Marti A. Hearst. 2021. Augmenting Scientific Papers with Just-in-Time, Position-Sensitive Definitions of Terms and Symbols. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) *(CHI '21)*. Association for Computing Machinery, New York, NY, USA, Article 413, 18 pages. doi:10.1145/3411764.3445648

[10] Jeffrey Heer, Matthew Conlen, Vishal Devireddy, Tu Nguyen, and Joshua Horowitz. 2023. Living Papers: A Language Toolkit for Augmented Scholarly Communication. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) *(UIST '23)*. Association for Computing Machinery, New York, NY, USA, Article 42, 13 pages. doi:10.1145/3586183.3606791

[11] Hyeonsu Kang, Joseph Chee Chang, Yongsung Kim, and Aniket Kittur. 2022. Threddy: An Interactive System for Personalized Thread-based Exploration and Organization of Scientific Literature. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (Bend, OR, USA) *(UIST '22)*. Association for Computing Machinery, New York, NY, USA, Article 94, 15 pages. doi:10.1145/3526113.3545660

[12] Hyeonsu B Kang, Nouran Soliman, Matt Latzke, Joseph Chee Chang, and Jonathan Bragg. 2023. ComLittee: Literature Discovery with Personal Elected Author Committees. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) *(CHI '23)*. Association for Computing Machinery, New York, NY, USA, Article 738, 20 pages. doi:10.1145/3544548.3581371

[13] Hyeonsu B Kang, Tongshuang Wu, Joseph Chee Chang, and Aniket Kittur. 2023. Synergi: A Mixed-Initiative System for Scholarly Synthesis and Sensemaking. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) *(UIST '23)*. Association for Computing Machinery, New York, NY, USA, Article 43, 19 pages. doi:10.1145/3586183.3606759

[14] J. Karat and C. M. Karat. 2003. The evolution of user-centered focus in the human-computer interaction field. *IBM Syst. J.* 42, 4 (Oct. 2003), 532–541. doi:10.1147/sj.424.0532

[15] Philippe Laban, Rania Hady, Dan Jurafsky, et al. 2025. How LLMs Hallucinate in Multi-Document Summarization. In *Findings of the North American Chapter of the Association for Computational Linguistics (NAACL Findings)*. https://aclanthology.org/2025.findings-naacl.293.pdf

[16] Yoonjoo Lee, Hyeonsu B. Kang, Matt Latzke, Juho Kim, Jonathan Bragg, Joseph Chee Chang, and Pao Siangliulue. 2024. PaperWeaver: Enriching Topical Paper Alerts by Contextualizing Recommended Papers with User-collected Papers. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 422, 19 pages. doi:10.1145/3613904.3642196

[17] Kyle Lo, Lucy Chang, Ashwin Paranjape, et al. 2024. The Semantic Reader Project: Augmenting Scholarly Documents through AI-Powered Interactive Reading Interfaces. *Commun. ACM* 67, 10 (2024), 72–81. doi:10.1145/3659096

[18] Catherine C. Marshall. 2003. Reading and Interactivity in the Digital Library: Creating an experience that transcends paper. https://api.semanticscholar.org/CorpusID:473212

[19] Phong Hai Nguyen, Kai Xu, Ashley Wheat, B. L. William Wong, Simon Attfield, and Bob Fields. 2016. SensePath: Understanding the Sensemaking Process Through Analytic Provenance. *IEEE Transactions on Visualization and Computer Graphics* 22 (2016), 41–50. https://api.semanticscholar.org/CorpusID:8838367

[20] Spyridon Papadopoulos, Vaibhav Agarwal, and Ravi Sharma. 2025. The Use of Generative Artificial Intelligence (AI) in Academic Research: A Review of the Consensus App. e77230 pages. doi:10.7759/cureus.77230

[21] Peter Pirolli and Stuart Card. 2005. The sensemaking process and leverage points for analyst technology as identified through cognitive task analysis. In *Proceedings of international conference on intelligence analysis*, Vol. 5. McLean, VA, USA, 2–4.

[22] Antoine Ponsard, Francisco Escalona, and Tamara Munzner. 2016. PaperQuest: A Visualization Tool to Support Literature Review. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (San Jose, California, USA) *(CHI EA '16)*. Association for Computing Machinery, New York, NY, USA, 2264–2271. doi:10.1145/2851581.2892334

[23] Nattakarn Rachatasumrit, Jonathan Bragg, Amy X. Zhang, and Daniel S. Weld. 2022. CiteRead: Integrating Localized Citation Contexts into Scientific Paper Reading. In *Proceedings of the 27th International Conference on Intelligent User Interfaces (IUI '22)*. ACM, 746–758. doi:10.1145/3490099.3511162

[24] Pushpakumar Ranganathan, Karun Sanjaya, S. Rathika, Ahmed Alawadi, Khamdamova Makhzuna, S. Venkatesh, and B. Rajalakshmi. 2023. Human-Computer Interaction: Enhancing User Experience in Interactive Systems. *E3S Web of Conferences* 399 (07 2023). doi:10.1051/e3sconf/202339904037

[25] Shirley Ann Rose. 1993. What's Love Got to Do with It? Scholarly Citation Practices as Courtship Rituals. *Language and Learning Across the Disciplines* 1, 3 (1993), 34–48.

[26] Trinita Roy, Asheesh Kumar, Daksh Raghuvanshi, Siddhant Jain, Goutham Vignesh, Kartik Shinde, and Rohan Tondulkar. 2024. SciSpace copilot: empowering researchers through intelligent reading assistance. In *Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence and Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence and Fourteenth Symposium on Educational Advances in Artificial Intelligence (AAAI'24/IAAI'24/EAAI'24)*. AAAI Press, Article 2855, 3 pages. doi:10.1609/aaai.v38i21.30578

[27] Bill N. Schilit, Gene Golovchinsky, and Morgan N. Price. 1998. Beyond paper: supporting active reading with free form digital ink annotations. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Los Angeles, California, USA) *(CHI '98)*. ACM Press/Addison-Wesley Publishing Co., USA, 249–256. doi:10.1145/274644.274680

[28] Bill N. Schilit, Morgan N. Price, and Gene Golovchinsky. 1998. Digital library information appliances. In *Digital library*. https://api.semanticscholar.org/CorpusID:14399540

[29] Cornelia Schoor, Sarah Hahnel, Nele Mahlow, Tamara Krstić, Iris Greving, Sarah Kuhfuss, and Maik Beege. 2022. Effects of Context and Discrepancy When Reading Multiple Documents. *Reading and Writing* 35, 6 (June 2022), 1449–1479. doi:10.1007/s11145-022-10321-2

[30] Craig S. Tashman and W. Keith Edwards. 2011. LiquidText: a flexible, multitouch environment to support active reading. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vancouver, BC, Canada) *(CHI '11)*. Association for Computing Machinery, New York, NY, USA, 3285–3294. doi:10.1145/1978942.1979430

[31] Nees Jan Van Eck and Ludo Waltman. 2010. Software Survey: VOSviewer, a Computer Program for Bibliometric Mapping. *Scientometrics* 84, 2 (2010), 523–538. doi:10.1007/s11192-009-0146-3

[32] Nees Jan Van Eck and Ludo Waltman. 2014. CitNetExplorer: A New Software Tool for Analyzing and Visualizing Citation Networks. *Journal of Informetrics* 8, 4 (2014), 802–823. doi:10.1016/j.joi.2014.07.006

[33] Jörg Zumbach and Maryam Mohraz. 2008. Cognitive Load in Hypermedia Reading Comprehension: Influence of Text Type and Linearity. *Computers in Human Behavior* 24, 3 (2008), 875–887. doi:10.1016/j.chb.2007.02.015